Band-1 Computational Experiment: Landau's Fourth Problem — Finite Enumeration of Primes n²+1

Human + ZoraASI full-dash (executor)

2026-09-11

Band-1 Computational Experiment: Landau-4 (Finite Enumeration of Primes $n^2 + 1$)

Honesty stamp. This note **enumerates** integers n with $1 \le n \le N$ such that $n^2 + 1$ is prime, for an explicit finite N computed on this box. It does **not** prove that there are infinitely many such primes. Finite enumeration $\neq$ infinitude. **Landau's fourth problem remains OPEN.** Iwaniec's 1978 theorem that $n^2 + 1$ is infinitely often a P_2 almost-prime is **not** a resolution of Landau-4. This is **not** a Clay Millennium claim and **not** a ToE / GATE-04 unlock.

Abstract

We scan $n = 1, \ldots, N$ and test primality of $n^2 + 1$ using deterministic-strength tests (`gmpy2.is_prime`, Baillie–PSW), exploiting that odd $n > 1$ make $n^2 + 1$ even and composite. We obtain

$$\mathbf{N = 10{,}000{,}000 = 10^7}, \qquad \mathbf{count = 456{,}362}.$$

Wall time for the primary enumeration: **2.57 s**. The OEIS A005574 prefix $(1, 2, 4, 6, 10, 14, \ldots)$ matches. Hardy–Littlewood / Landau–Shanks heuristic ratio (count/HL ≈ 1.07) (heuristic agreement is **not** a proof). **Infinitude status: OPEN.**

Problem statement

Landau's fourth problem (ICM 1912). Are there infinitely many primes of the form

$$p = n^2 + 1, \qquad n \in \mathbb{Z}_{\ge 1}?$$

This experiment. Enumerate all such $n \le N$ for a chosen explicit N tractable on this box (seconds), list them, count them, and record the literature ledger — especially Iwaniec's almost-prime result — without elevating almost-primes to primes.

Literature ledger (not re-proved on box)

Year	Result	Authors	Status
1912	Listed among Landau's four problems	Landau (ICM)	historical framing
1978	Infinitely many n with $\Omega(n^2 + 1) \le 2$ (P_2)	**Iwaniec**, *Invent. Math.* **47**	**proved almost-prime — NOT primes**

Year	Result	Authors	Status
1952+	Under ERH: primes $x^2 + y^2$ with $y = O(\log p)$	Ankeny; Kubilius	**CONDITIONAL** thin-set — **not** $y = 1$
—	Unconditional near-square $y = O(p^{0.119})$	Harman–Lewis	thin cousin — still not $y = 1$
—	HL / Bateman–Horn $\sim C \int dn/\log(n^2+1)$, $C \approx 1.3728$	Hardy–Littlewood; Landau–Shanks	**heuristic only**

Primary almost-prime citation (careful: **not** full resolution): H. Iwaniec, *Almost-primes represented by quadratic polynomials*, Inventiones Mathematicae **47** (1978), 171–188 (DOI 10.1007/BF01578070). This proves infinitely many n for which $n^2 + 1$ has **at most two** prime factors. That is **strictly weaker** than infinitely many primes $n^2 + 1$.

No standard theorem asserts GRH ⇒ Landau-4. Conditionally thin near-square results (Ankeny–Kubilius under ERH) do **not** settle the $y = 1$ case.

Unconditional infinitude of primes $n^2 + 1$ remains OPEN.

Method (box)

1. **Unit tests.** OEIS A005574 prefix; $1^2 + 1 = 2$; $2^2 + 1 = 5$; odd $n = 3$ composite; `gmpy2` prime/composite sanity.
2. **Parity filter.** For odd $n > 1$, $n^2 + 1$ is even and > 2 — discard without a primality test.
3. **Primality.** For $n = 1$ and even $n \le N$, test $n^2 + 1$ with `gmpy2.is_prime` (BPSW).
4. **Outputs.** Full CSV list of (n, p); decade cumulative counts; HL ratio diagnostic.

Not used as a claim surface: Iwaniec P_2 as "Landau-4 solved"; GRH⇒Landau-4; preprint "resolutions"; Clay / ToE / GATE-04.

Results (numbers!)

Unit tests

`PASS`: OEIS A005574 prefix; $n = 1, 2$; odd-n composite; `gmpy2` sanity.

Box enumeration

Quantity	Value
Primary N	$10{,}000{,}000 = 10^7$
Integers n scanned (parity-filtered)	$1 + 5{,}000{,}000$ candidates
Count ($n \le N$, $n^2 + 1$ prime)	456,362
Largest n	9,999,960
Largest $p = n^2 + 1$	99,999,200,001,601
HL estimate $\approx C\,N/(2\log N)$	$\approx 425{,}860.94$
Count / HL	≈ 1.0716
Primary wall	**2.5730 s**
OEIS A005574 prefix	**PASS**

Cumulative counts by decade ($n \le 10^k$)

k	10^k	count
1	10	5
2	100	19
3	1{,}000	112
4	10{,}000	841
5	100{,}000	6{,}656
6	1{,}000{,}000	54{,}110
7	10{,}000{,}000	**456{,}362**

Sample hits (first 15)

n	$p = n^2 + 1$
1	2
2	5
4	17
6	37
10	101
14	197
16	257
20	401
24	577
26	677
36	1{,}297
40	1{,}601
54	2{,}917
56	3{,}137
66	4{,}357

Full list of all **456{,}362** pairs: `primes_n2_plus_1_list.csv` in the experiment directory.

What this does not claim

- Infinitely many primes $n^2 + 1$ (still **OPEN**).
- That Iwaniec's P_2 theorem resolves Landau-4.
- GRH $\Rightarrow$ Landau-4 (no standard theorem).
- Clay Millennium / ToE / GATE-04 unlock.
- That heuristic HL agreement proves infinitude.

Reproducibility

```
python3 experiments/band1/landau_n2_plus_1/landau_n2_plus_1_enumerate.py
# writes results.json, literature_ledger.json, SUMMARY.txt, primes_n2_plus_1_list.csv
```

Requires: Python 3, `gmpy2`.

Conclusion

For $n = 1, \ldots, 10^7$, there are **456{,}362** primes of the form $n^2 + 1$ (wall ~ 2.6 s; OEIS prefix match). Iwaniec (1978) gives infinitely many P_2 almost-primes — carefully **not** infinitude of primes. **Landau's fourth problem remains OPEN.**